

What lame super power
would you rather have?

CS 340

Concurrency | OS

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Q1

~Code~
340

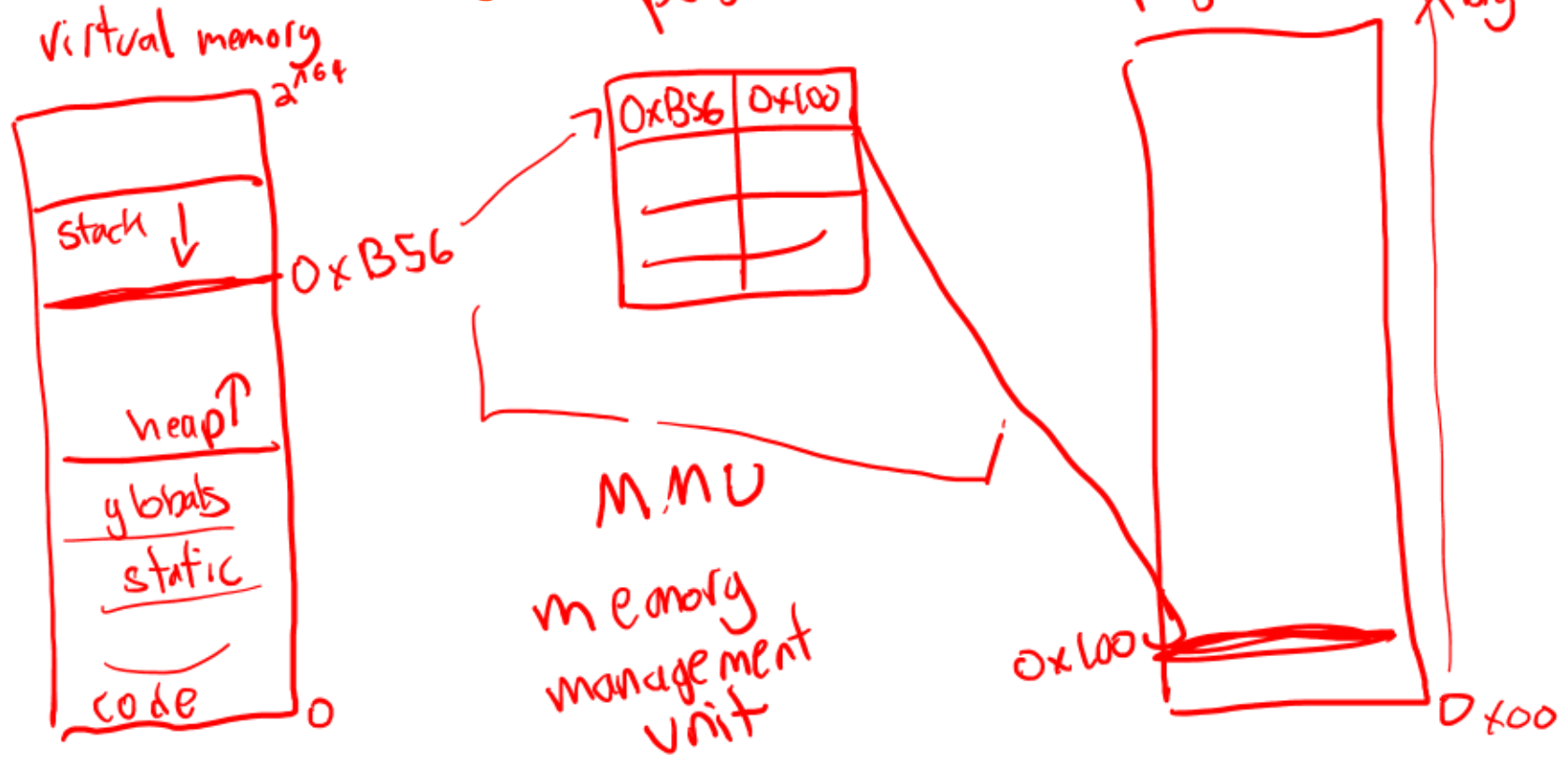


Updates

1. MP4 Allocator due Tuesday
2. Exam 1 - Retake Grades Released
 - a. Regrade requests
 - i. By next week Thursday
 - ii. Submit an issue on Prairie Learn
3. Exam 2 sign up open
4. HW 6 out now, due next Wednesday

upload grades
to canvas

Virtual Memory



Virtual Memory Takeaways

- Virtual memory - isolates programs for security and the illusion of space.
- You (the programmer) cannot see or know what physical address your code is using. You can only access the virtual addresses for your specific program.
- Each running program gets its own page table and virtual memory space. Virtual addresses may overlap between programs but physical addresses cannot.
- **Physical memory for a program can vary in size and location.**
- Virtual memory has a fixed size and a consistent location.



Am I guaranteed that ptr and ptr2 hold values 6,000 bytes apart in physical memory?

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Q2

~Code~
340



char x = 7;

char *ptr = (&x);

char *ptr2 = ptr + 6000;



Virtual Memory Takeaways

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- Each running program gets its own page table and virtual memory space. Virtual addresses may overlap between programs but physical addresses cannot.
- Physical memory for a program can vary in size and location.
- **Virtual memory has a fixed size and a consistent location.**

Am I guaranteed that ptr and ptr2 hold values 6,000 bytes apart in virtual memory?

```
char x = 7;
```

```
char *ptr = &x;
```

```
char *ptr2 = ptr + 6000;
```

yes

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Q3

~Code~
340

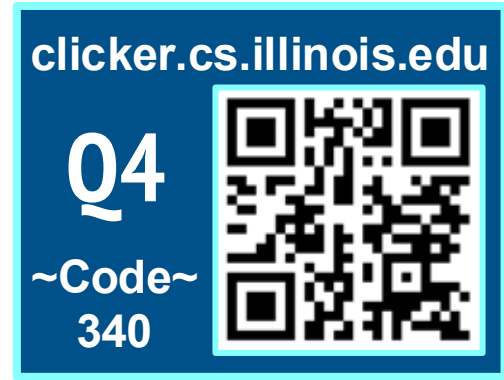


Virtual Memory Takeaways

- Virtual memory - isolates programs for security and the illusion of space.
- You (the programmer) cannot see or know what physical address your code is using. You can only access the virtual addresses for your specific program.
- Each running program gets its own page table and virtual memory space. Virtual addresses may overlap between programs but physical addresses cannot.
- Physical memory for a program can vary in size and location.
- Virtual memory has a fixed size and a consistent location.

If program 1 stores my password in a c-string at virtual address 0xB50632 will this code in program 2 get me that password?

```
char *ptr = (char*)0xB50632;  
printf("%s", ptr);
```



no

Virtual Memory Takeaways

- Virtual memory - isolates programs for security and the illusion of space.
-  You (the programmer) cannot see or know what physical address your code is using. You can only access the virtual addresses for your specific program.
- Each running program gets its own page table and virtual memory space. Virtual addresses may overlap between programs but physical addresses cannot.
- Physical memory for a program can vary in size and location.
- Virtual memory has a fixed size and a consistent location.

Is it possible for these two programs to print out the same value while running at the same time?

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Q5

~Code~
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```
int x = 5;  
printf("%#x", &x);
```

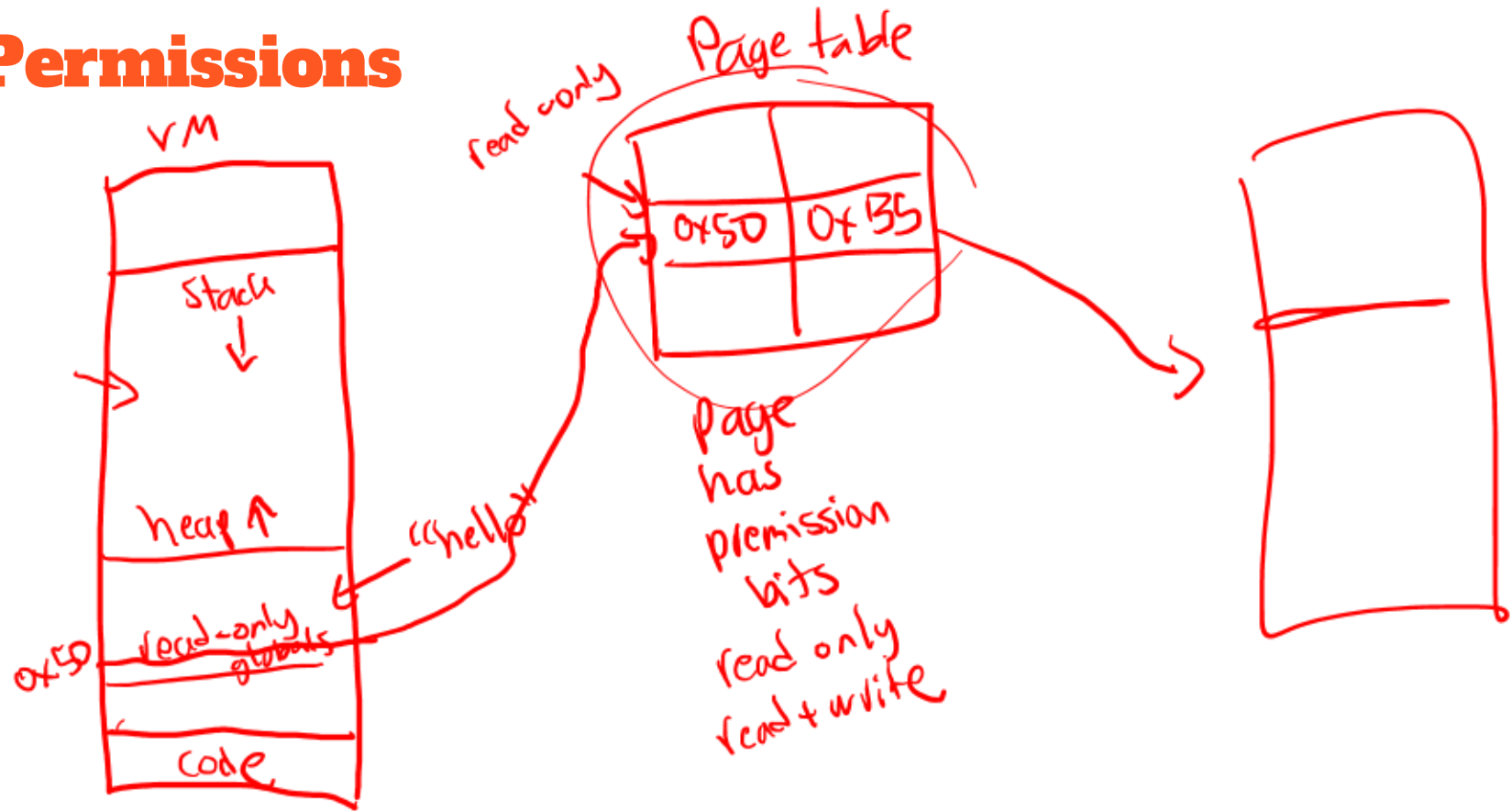
A red arrow points to the format string "%#x" in the second line, and a red circle highlights the "&x" argument.

```
int x = 2;  
printf("%#x", &x);
```

A red circle highlights the "&x" argument in the second line.

yes

Permissions



Concurrency | OS

Today's LGs - Expand your mental model of a computer to include running multiple program concurrently.

Be able to answer conceptual questions about concurrency and the Operating System (OS)

Agenda

1. Operating System (OS)
1. Concurrency with Processes
1. Practice Clickers

The Operating System and Kernel



OS - Software that manages the hardware and shares it among running programs. Includes the kernel, system utilities, device drivers, libraries, and user interfaces.

Kernel - Part of the OS with special privileges to manage programs and hardware

Kernel

What does the Kernel do?

-manages machine's resources while keeping programs isolated

Memory - through virtual memory

Storage, devices - hardware only accessed through the kernel

Processes - who runs when

↓
running instance
of a program

CPU Privilege Modes

Two modes: *user, kernel*

Privileged instructions (e.g., changing page tables, accessing devices) only runs in *Kernel* mode.

CPU enters *kernel* mode on a -

- System call - the program asks the kernel for a service
like file reading
- Interrupt - hardware need attention - unrelated to running program
- Exception - instruction can't complete
kernel takes over and decides what to do

OS - Kernel : Big Picture

Review - What is true about a CPU?

- ☒ A) It does fetch, decode, and execute cycles
- ☒ B) It can decode and execute basic instructions like jump, load, store, arithmetic → per ISA
- ☒ C) It can decode and execute C code directly → compiled only
- ☒ D) It uses gates to implement logic

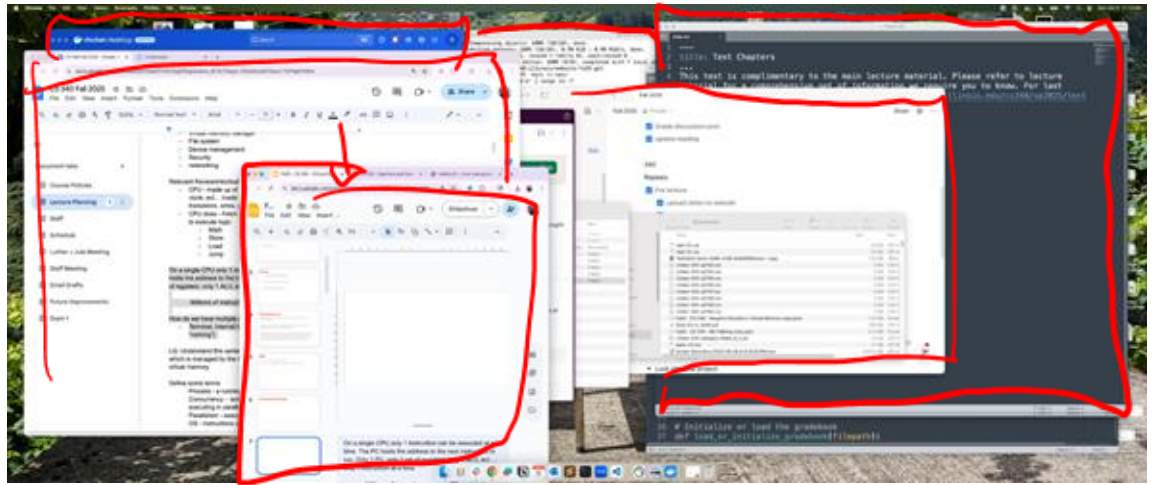
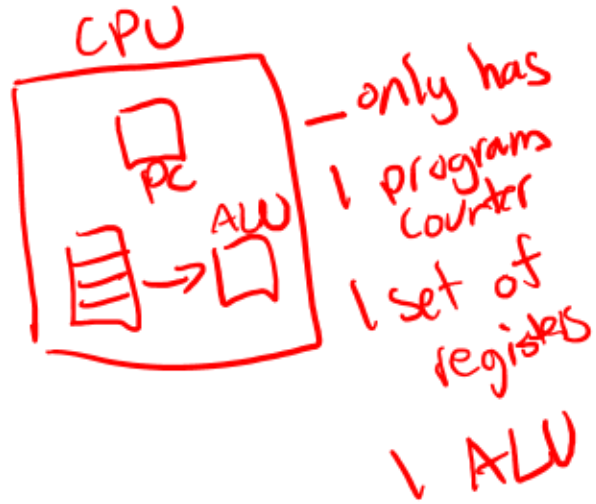
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How?



LG - Be able to explain each part of this sentence

Processes are running concurrently which is managed by the OS and is safe because of kernel mode and virtual memory.

Vocabulary

Process - a running instance of a program

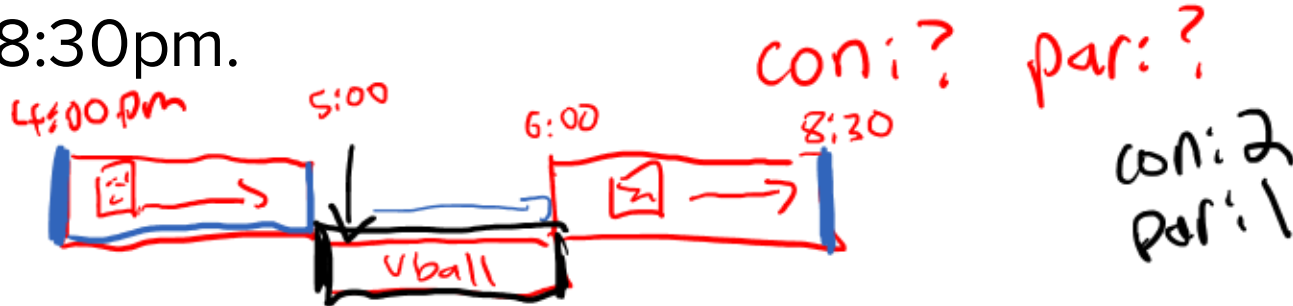
Concurrency - active at the same time

Parallelism - executing at the same time



At 5:15pm how many activities am I doing concurrently? In parallel?

I start my stats HW at 4:00pm, at 5:00pm I take a break start a game of volleyball which ends at 6:00pm, I then work on and finish my stats HW at 8:30pm.



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Q7

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At 7:00pm how many activities am I doing concurrently? In parallel?

I start my stats HW at 4:00pm, at 5:00pm I take a break start a game of volleyball which ends at 6:00pm, I then work on and finish my stats HW at 8:30pm.

~~con~~
con: 1 par: 1

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Q8

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LG - Be able to explain each part of this sentence

Processes are running concurrently which is managed by the OS and is safe because of kernel mode and virtual memory.

Processes are running concurrently



Only 1 runs at a time

Kernel → switches processes in and out of the CPU very quickly

**What is the closest estimate
to how many instructions a
CPU can run in a second?**

1,000,000,000

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Q9

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If I'm typing at 200 characters a minute, what is the minimum number of CPUs I need to seamlessly handle updating my notes app while running my MP1.2?

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Q10

~Code~
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1 - cpu
↓
1,000,000,000 per second

The Operating System - Kernel

manages

-memory ← virtual memory

-Storage devices

- Processes - who runs when

- creates, destroys, schedules

- Context Switch

- saving current processes progress in
PCB

- load in another processes progress

Processes are running concurrently which is managed by the OS

- Every few milliseconds timer interrupt ↙ system call exception
- Hardware in the CPU switches to kernel mode
 - load in kernel code
 - to - some tasks, → requesting a file from disk
 - potentially context switch

fopen() Example

program.c
fopen("image.png", "r");

- instruction in the cpu for program.c ← compiled version that is system call to request a file
- Triggers kernel mode
- kernel request those bytes from disk
- kernel context switch in another process → starwren valley
- file comes back from disk → trigger on interrupt
- kernel mode
- context switch back to program.c with bytes of image.png

LG - Be able to explain each part of this sentence

Processes are running concurrently which is managed by the OS and is safe because of ***kernel mode*** and ***virtual memory***.

Is it possible for these two programs to print out the same value while running concurrently?

yes

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Q11

~Code~
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```
int x = 5;  
printf("%#x", &x);
```

```
int x = 2;  
printf("%#x", &x);
```

Is it possible for these two programs to be running concurrently? Given the assembly code the C code compiled into?

can they be using the same register?

```
int x = 5;
```

```
x = x + 1;
```

→ (load x into reg1)

(add reg1 with 1 and store in reg1)

```
int x = 2;
```

```
x = x + 1;
```

→ (load x into reg1)

(add reg1 with 1 and store in reg1)

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Q12

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yes

What does the OS not do?

↓
code

- ~~A) Does not schedule processes~~
- ☒ B) Does not execute user and kernel instructions ?
- ~~C) Does not manage virtual memory~~
- D) Does all of the above

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Q13

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What manages the concurrent processes on your computer?

- A) CPU
- ☒ B) OS
- C) ALU
- D) IDE

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Q14

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Can a user process search through and edit anything on your computer? Why or why not?

no

virtual memory → go through page table → kernel to get the memory

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Q15

~Code~
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How do the following connect or relate to each other?

Transistors, gates, MUX, CPU, processes, kernel mode, virtual memory, and the OS



LG - Be able to explain each part of this sentence

Processes are running concurrently which is managed by the OS and is safe because of kernel mode and virtual memory.